

Week 5

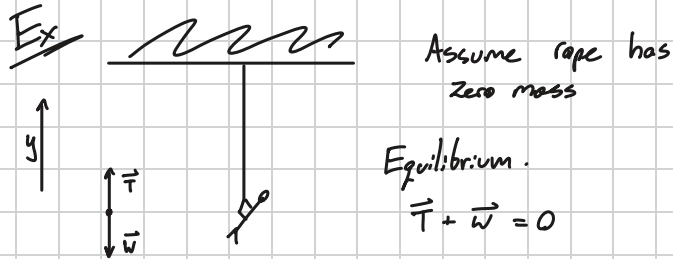
Newton's first Law

→ Particle in equilibrium

Condition for equilibrium $\sum \vec{F} = 0$

in 2D $x: \sum F_x = 0$

$y: \sum F_y = 0$



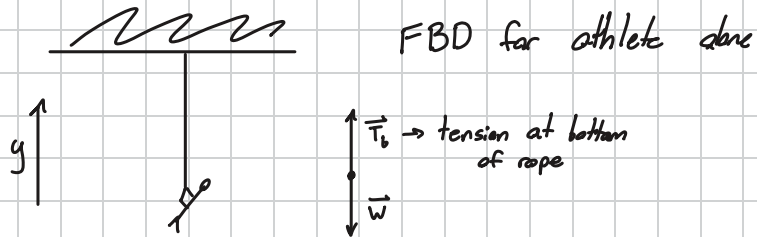
Equilibrium.

$$\vec{T} + \vec{W} = 0$$

$y: W = -mg$

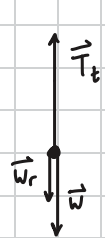
$$T - mg = 0 \Rightarrow T = mg$$

~~Ex~~ rope has mass m_r



$$T_b - mg = 0 \Rightarrow T_b = mg$$

FBD for athlete + rope system



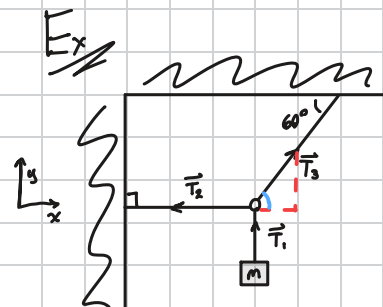
$\vec{T}_t$ - tension at top of rope

$$\vec{T}_t + \vec{w} + \vec{w}_r = 0$$

$$w = -mg, w_r = -m_r g$$

$$\Rightarrow T_t - mg - m_r g = 0 \Rightarrow$$

$$T_t = (m + m_r)g$$



the string and ring are massless. find tension.

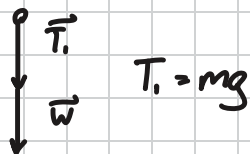
Forces on ring: $\vec{T}_1, \vec{T}_2, \vec{T}_3$

Ring is in equilibrium: $\vec{T}_1 + \vec{T}_2 + \vec{T}_3 = 0$

$$x: -T_2 + T_3 \cos \theta = 0 \quad (1)$$

$$y: -T_1 + T_3 \sin \theta = 0 \quad (2)$$

FBD for mass m:



$$T_1 = mg$$

Sub into (2): $-mg + T_3 \sin \theta = 0$

$$\Rightarrow T_3 = \frac{mg}{\sin \theta}$$

Now Sub into ①

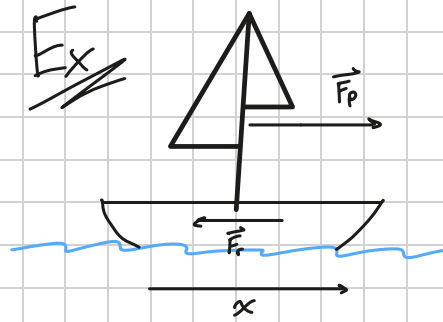
$$-T_2 + \frac{mg}{\sin\theta} \cdot \cos\theta = 0$$

$$\Rightarrow T_2 = \frac{mg \cos\theta}{\sin\theta} = mg \cot\theta$$

Newton's Second Law

$$\sum \vec{F} = m\vec{a}$$

$$\begin{aligned} \text{2D problem} \Rightarrow \quad X: \sum F_x &= ma_x \\ Y: \sum F_y &= ma_y \end{aligned}$$



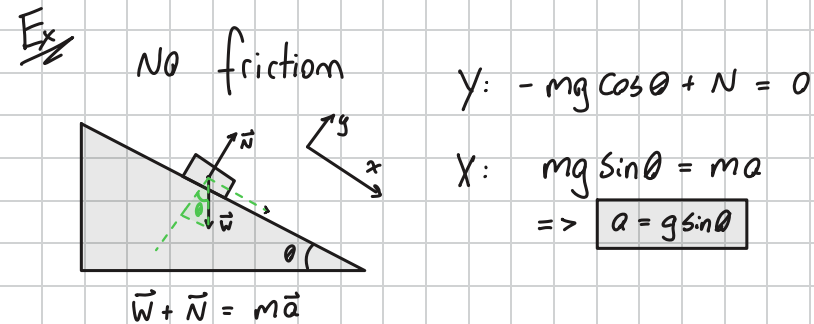
$$M = 200 \text{ kg}$$

$$F_r = 100 \text{ N}$$

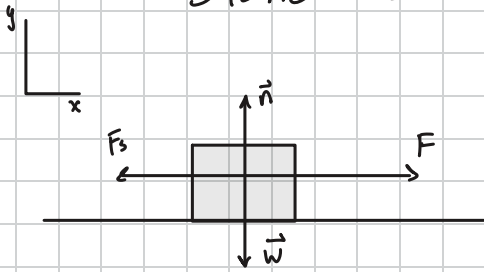
$$a_x = 1.5 \text{ m/s}^2$$

What is F_p ?

$$X: F_p - F_r = ma_x \Rightarrow F_p = F_r + ma_x \Rightarrow F_p = 400 \text{ N}$$



Static vs kinetic friction



no motion

$$y: \vec{n} + \vec{w} = 0$$

$$n - mg = 0$$

$$\Rightarrow n = mg$$

$$x: \vec{F} + \vec{F}_s = 0$$

$$x: F - F_s = 0 \quad \Rightarrow F_s = F$$

there is a maximum friction force

$$F_{smax} = \mu_s n \quad \Rightarrow F_s \leq \mu_s n$$

if $F > F_{smax} \Rightarrow$ object starts moving

$$F_k = \mu_k n$$

in practice $\mu_k < \mu_s$

Example

A crate weighs 500N

A force of 200N is needed for const vel

with initl of 230N needed

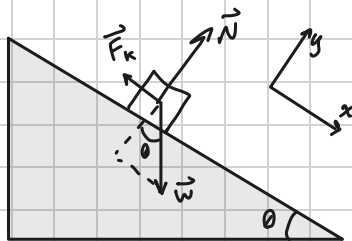
$$F_{s, \max} = \mu_s \cdot mg = 230 \text{ N}$$

$$\mu_s = \frac{230 \text{ N}}{500 \text{ N}} = 0.460$$

$$\text{Once moving: } F_k = \mu_k \cdot mg = 200$$

$$\Rightarrow \mu_k = \frac{200 \text{ N}}{500 \text{ N}} = 0.400$$

E_x



$$W_x = W \sin \theta$$

$$W_y = W \cos \theta$$

$$W_x = mg \sin \theta$$

$$W_y = -mg \cos \theta$$

Newton's Second Law:

$$\vec{W} + \vec{n} + \vec{F}_k = m\vec{a}$$

$$x: mg \sin \theta - \mu_k n = ma$$

$$y: n - mg \cos \theta = 0 \Rightarrow n = mg \cos \theta$$

$$\Rightarrow mg \sin \theta - \mu_k mg \cos \theta = ma$$

$$\cancel{m}g(\sin \theta - \mu_k \cos \theta) = \cancel{m}a$$

$$\Rightarrow a = g(\sin \theta - \mu_k \cos \theta)$$

Note: formula only works

$$\text{if } a > 0 \quad (\sin \theta > \mu_k \cos \theta)$$

$$\text{or } \mu_k < \frac{\sin \theta}{\cos \theta} = \tan \theta$$

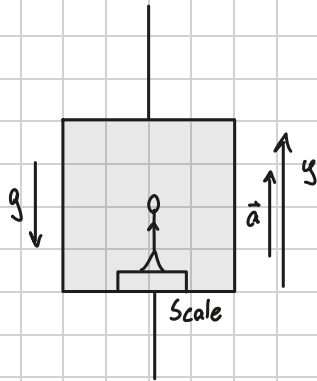
if $a = 0$

$$a = g(\sin\theta - \mu_k \cos\theta) = 0$$

$$\Rightarrow \mu_k = \tan\theta$$

Ex

Apparent weight in an elevator



$$\vec{n} + \vec{w} = m\vec{a}$$

$$n - mg = ma$$

$$\Rightarrow n = mg + ma = m(g + a)$$

$$a > 0 \Rightarrow n > mg$$

Note: the scale is measuring the normal force

